PSYC1071: Introduction to Psychology 1
Animal Learning and Cognition

Lecture 9:
Language and communication

Professor Anthony McGregor

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Department of Psychology

- Communication in animals
  - Honeybees (Von Frisch)
  - Vervet monkeys (Seyfarth & Cheney)
- Can animals acquire language?
  - Language is unique to humans - Chomsky, Macphail
  - Hockett's criteria for language
  - Attempts to train apes to use language
    - Washoe
    - Sarah
    - Kanzi
    - Nim
  - Is it language?
  - Other studies - Parrot, Dolphins

Communication and Language

Communication occurs when one organism

transmits a signal to another organism that

is capable of responding appropriately to that signal

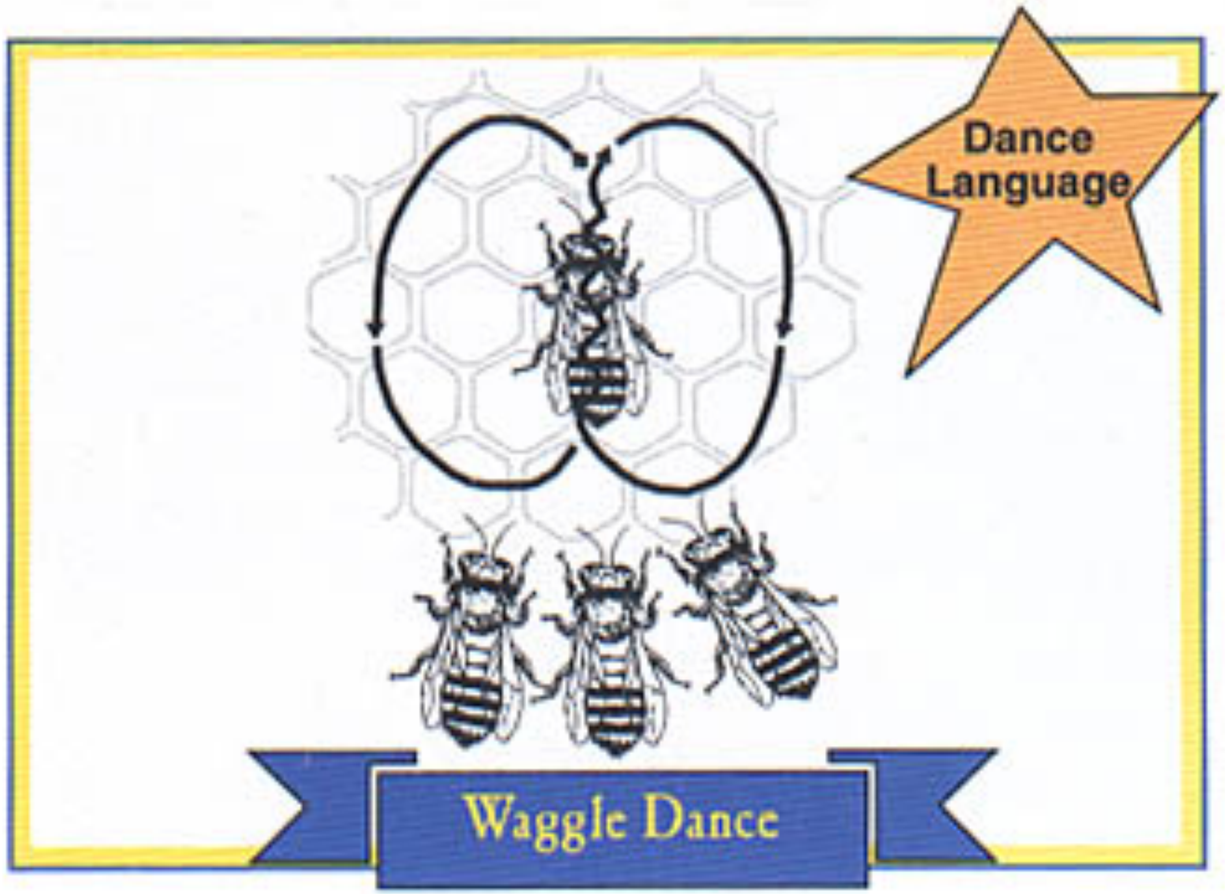
Communication and Language

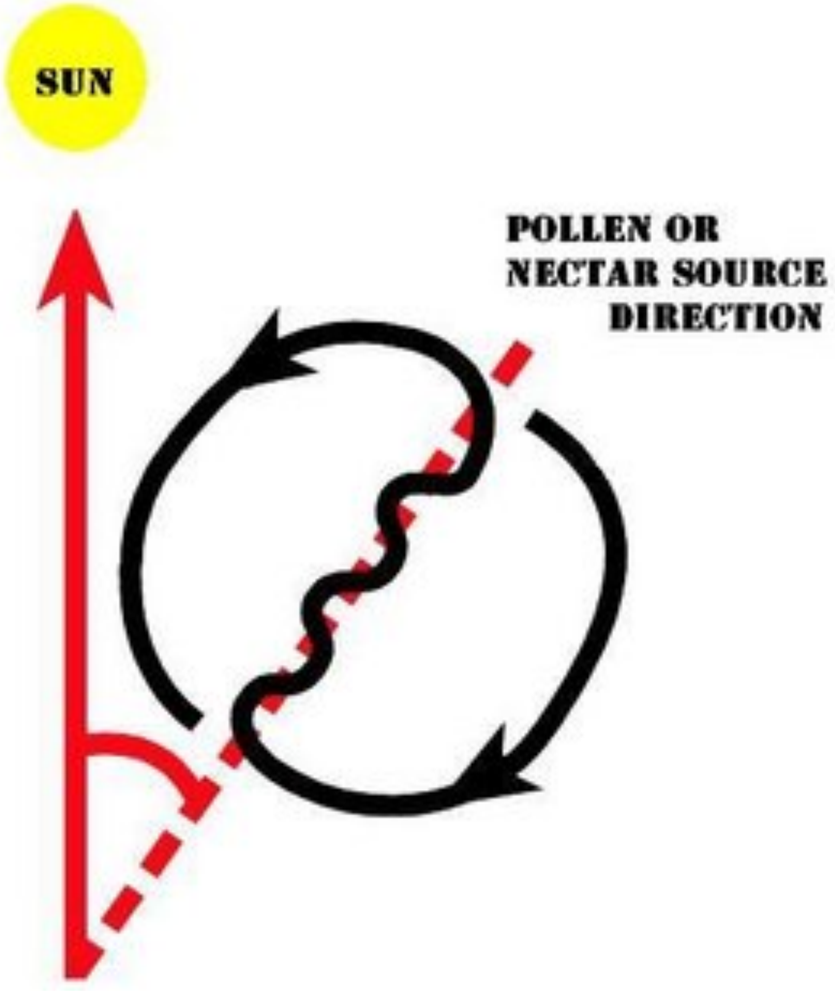
- Honey Bee
  - Von Frisch (1974)

Round Dance

50-100m of hive

> 100m from hive



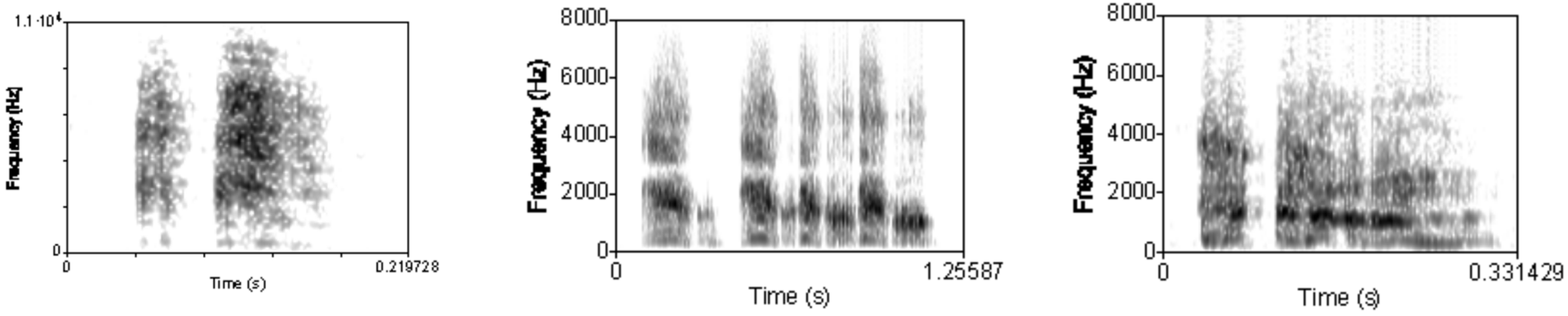

SUN
POLLEN OR
NECTAR SOURCE
DIRECTION

Communication and Language

- Honey Bee
  - Von Frisch (1974)

- Vervet Monkey
  - Seyfarth and Cheney (1993)

- Snake call
  - Look on ground
- Leopard call
  - Run to trees
- Eagle call
  - Look up to sky



Can Animals Acquire Language?

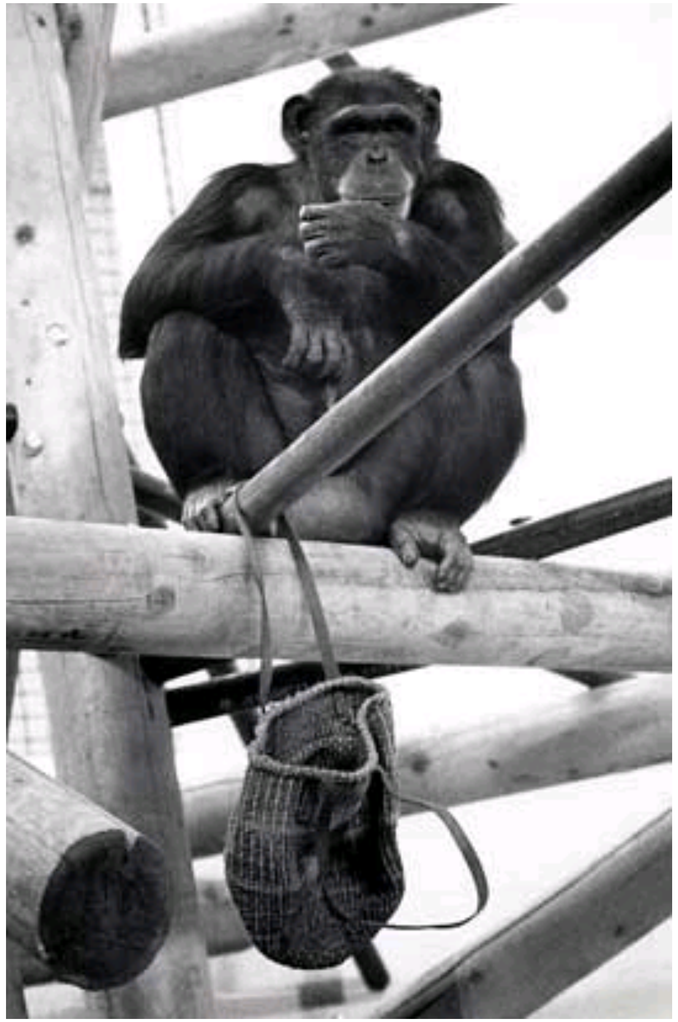
- Language is unique to humans
  - Chomsky
  - Macphail

What is Language?

- Hockett's criteria
  - Discrete units
  - Arbitrary units
  - Semanticity
  - Displacement
  - Productivity through syntax

Attempts to teach language to an ape

- Furness (1916) female orang-utan
- Hayes & Hayes (1951) Vicki, a chimp
- Gardner & Gardner (1969) Washoe



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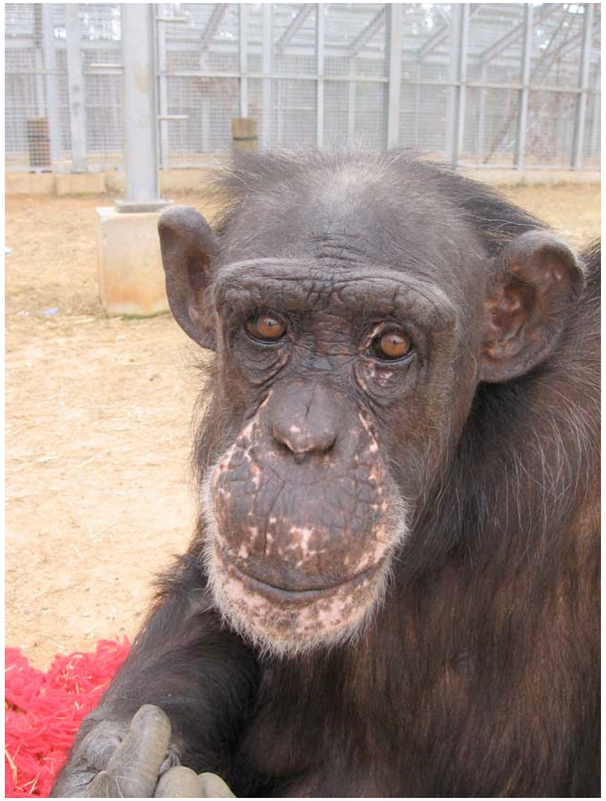

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- Rumbaugh & Savage-Rumbaugh

  Lana, Austin, Sherman, Kanzi

Lana

Sherman

Sue

I
Brush
Said
ed past
Ear
Can
Broken
Swelling
For
Sick
New
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Scent
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Fish
Break
Night
Clippers
Instrument
Uphill
Medicine
See
different Different
different Different
Write
Picture
Puzzle
DO
Stop
Mask
Here
P-Suke
Like
Ready
Chalk
Pretend
Close
Take
More
Fish
Need
Nyota
The
That
AY
Stick
Run
What ????
Wise
Time
Vegetable
Little
Down
It
Is
Space
Careful
Agree
Sit
And
There
Make
Sorry
Right
BE
Crayon
Feel
Lunch
Electric Shock
Put
Point
Home
Are
Show
Squirrel
Thing
Wait
This
Row
Neema
Night
Pour
Not
Listen
Visitors
Watermelon
Candy
Box
Was
Wrong
Then
S plural
Today
Where
Many
We
Before
ME
Blackboard
Back
Eat
Wait
Wait
Drink
If
RED
Place
Those
Bird
Color
Want
Cooler
Count
Yours
Piece
Trolley
Up
To
Try
Arm
Night
Blue
Have
My
You

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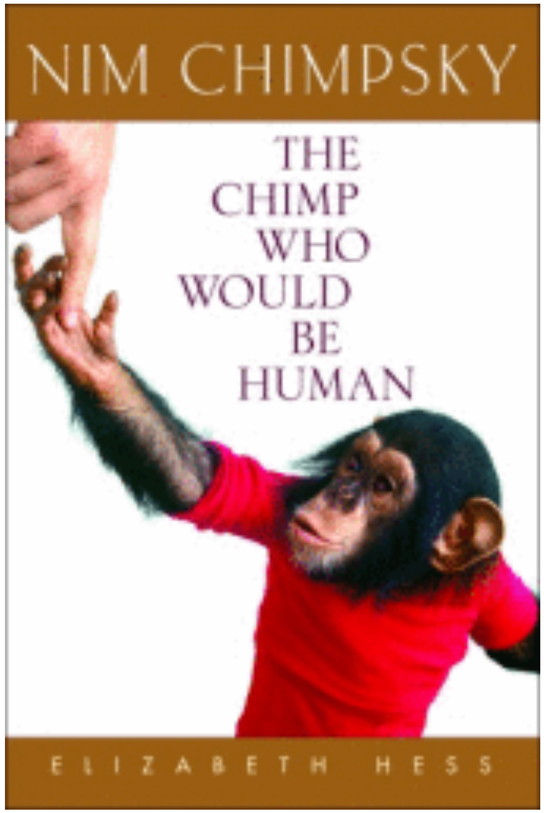
  Lana, Austin, Sherman, Kanzi
- Terrace (1979) Nim


NIM CHIMPSKY
THE CHIMP WHO WOULD BE HUMAN
ELIZABETH HESS

Can an ape create a sentence?

- Terrace et al. (1979) Nim Chimpsky
  - 19,000 utterances
  - 5,235 types

What is Language?

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Other Studies

- Pepperberg (1992) Alex, a parrot
- Herman et al. (1993) dolphins
- Savage-Rumbaugh (1994)
  - Kanzi

Rule:	Indirect object	Direct object	Action
Example:	Hoop	Pipe	Fetch

Interpretation: Take the pipe to the hoop

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PSYC1071 - Animal learning and cognition summary

Historical background

- Romanes-Lloyd Morgan-Thorndike-Watson-Skinner-Pavlov

Operant and Pavlovian conditioning

What is intelligence? How is it distributed?

Animal Memory

- Capacity-Timing-Number-Categories
- STM and LTM

Associative Learning

- What determines the nature of the CR?
- Conditions of learning - Contiguity, contingency, surprise, attention
- Human Pavlovian conditioning - counterconditioning, aversion therapy, phobia, desensitisation
- Human Instrumental conditioning - what's reinforcing, when to present a reinforcer

Problem-solving

- Trial and error or reasoning?
- Navigation - path integration-landmark use-cognitive map
- Insight - importance of prior experience

Communication and language

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- Can animals acquire language?